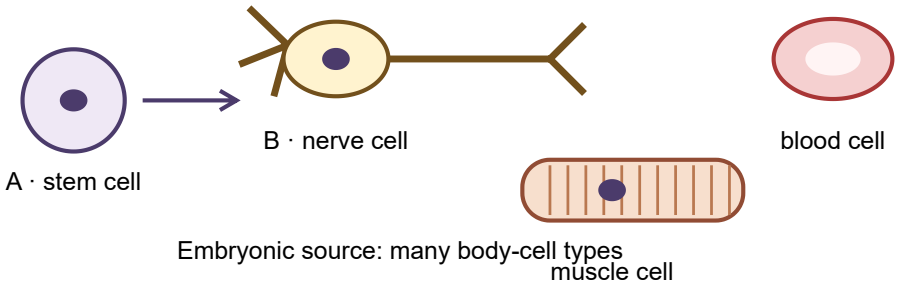
Reference A animal cells



An embryonic stem cell can give rise to many specialised body-cell types.

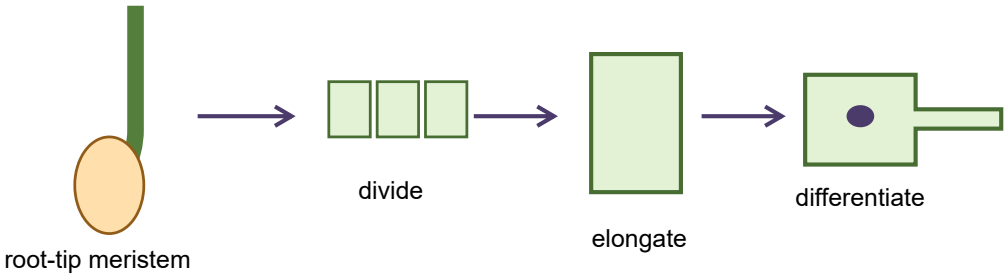
Reference only. Do not write on this sheet. Trace a feature, then explain its function.

Cell or source	Feature or range	Function or limit
Nerve cell	Long extension	Carries signals over a distance
Muscle cell	Structures that contract	Contraction produces force
Embryonic stem cell	Can give rise to many body-cell types	Not a promise of a successful treatment
Adult blood-forming stem cell	More limited range of blood-cell types	Makes replacement blood cells

The arrows show possible lineages. The drawing does not show every cell stage. Cells do not choose their own jobs. Specialised cells can share inherited DNA but use it differently.

Made by Matt

Reference B root growth



Root-tip meristem, division, elongation and differentiation in a simplified route.

Reference only. Do not write on this sheet.

Process or region	What it adds	Useful limit
Meristem	A region where cells divide	Not every root cell remains a meristem cell
Division	More cells	Number alone does not prove a specialised role
Elongation	Longer cells	Not the name for becoming specialised
Differentiation	Features suited to a job	Different cells can be at different stages

A root hair extension provides a large surface for absorption. The route is simplified. Use it to compare processes, not as a claim that every cell follows an identical sequence.

Made by Matt

Arrival and exit Independent

Use the labelled picture on Reference A for question 1. Respond in your usual mode.

1. Read the picture labels. Which letter labels the stem cell?

2. What does mitosis followed by cell splitting add to a tissue?

3. How does making more cells differ from a cell gaining a specialised job?

4. Where in a growing root are cells made by division?

Exit ticket

1. Complete: A cell becomes specialised by ___.

2. Name one plant growth process after division and say what it changes.

Pupil voice

What helped you most today, or what would you change next time?

Point, speak, sign, draw, write or choose. You may pass.

verbal · written · drawn · signed · symbol / choice-selected · demonstrated / re-attempt · adult-scribed. You may pass.

Made by Matt

You do Independent

Use your book or this sheet. One response per task. Use Reference A and B.

1. Explain the difference between division and differentiation. [2] Staff I ☐ S ☐ C ☐

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2. Use a labelled nerve-cell drawing to link one feature to its function. [3] Staff I ☐ S ☐ C ☐

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3. Compare the range of an embryonic stem cell with that of an adult blood-forming stem cell. State one limit of the model. [3] Staff I ☐ S ☐ C ☐

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4. Draw and label a simplified growth route at a root tip. Include division, elongation, differentiation and meristem. [4] Staff I ☐ S ☐ C ☐

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5. Compare growth and repair in animal skin with growth in a root. Give one shared process and one difference, with an explanation. [3] Staff I ☐ S ☐ C ☐

Feedback and response · Independent

Feedback codes

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt